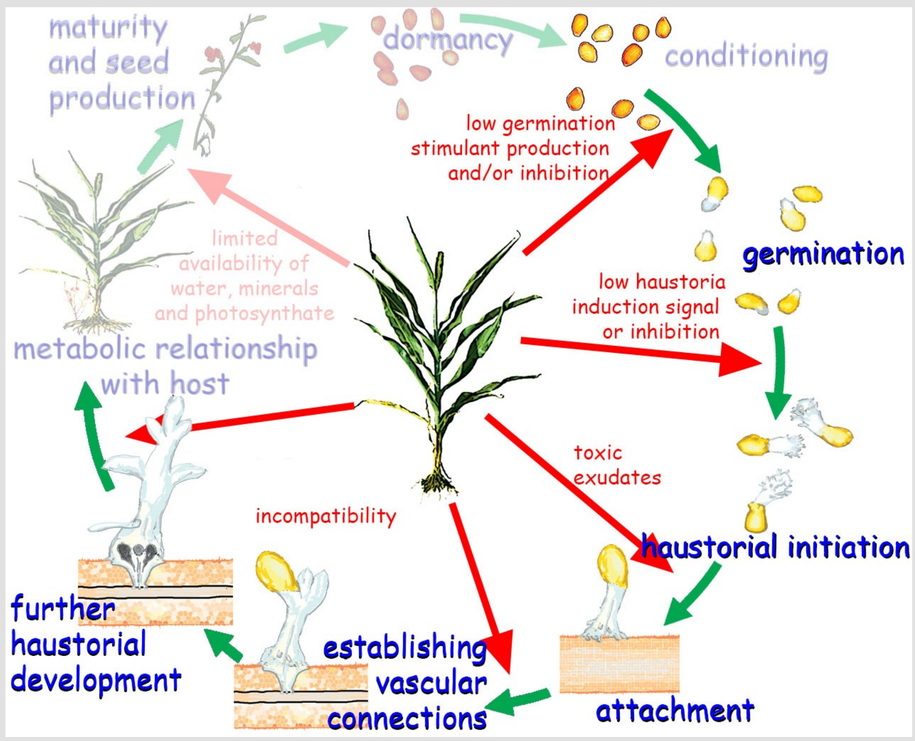
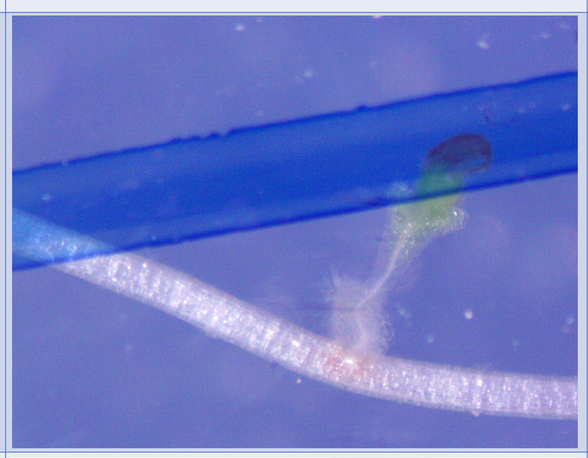
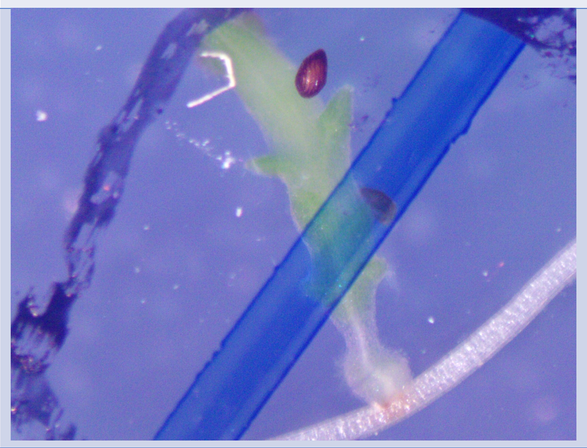
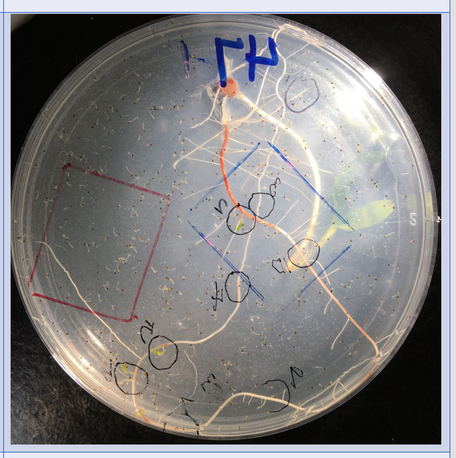
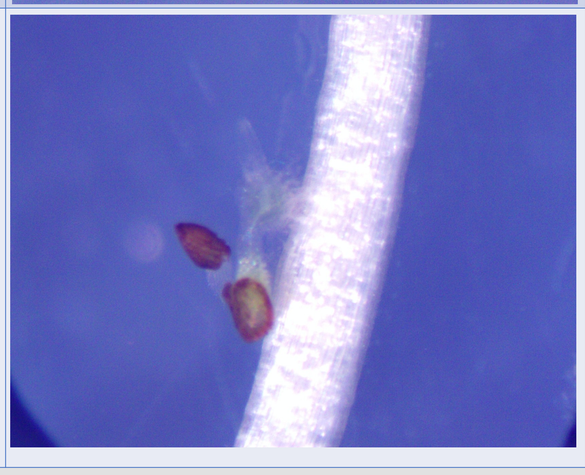
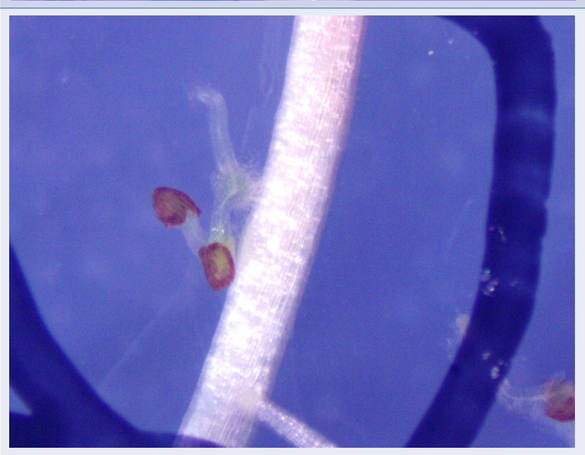
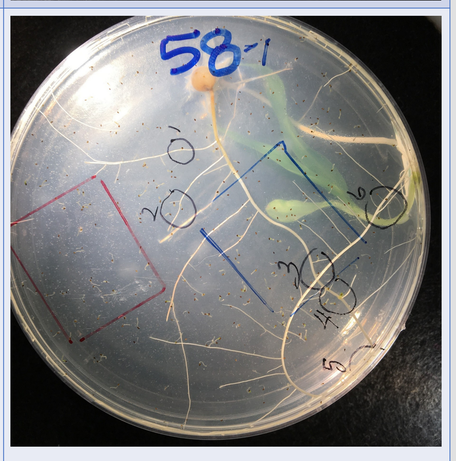
Laboratory screening of sorghum lines for incompatibility, a post-attachment resistance mechanism to the parasitic weed *Striga hermonthica*

Cameron Matthews, Adedayo Adeyanju, Patrick J. Rich, Nate Bowser, Gebisa Ejeta
Department of Agronomy, Purdue University, 915 W. State St., West Lafayette, IN 47907
matthe37@purdue.edu

Abstract:

Striga hermonthica is an obligate root parasitic weed in sorghum, reducing crop yield severely, especially in low input agriculture common across sub-Saharan Africa. Host plant resistance is an essential component of *Striga* control. Only one resistance gene is known, *LGS1*, controlling strigolactone production. Sorghum *lgs1* mutants have reduced *Striga* germination stimulant activity. After germinating in response to host exuded strigolactones, *Striga* attaches to its roots. Another resistance mechanism called incompatibility occurs at the post-attachment stage. *Striga* on an incompatible host fails to form the functional xylem connection necessary to sustain continued growth. We screened 58 inbred sorghum lines of common pedigree, including some released as resistant varieties in Africa, by laboratory co-culture that allowed successive observations of parasitic attachments. Differences in *Striga* germination stimulant activity between lines were overcome by exogenous application of strigolactones. One parent was the donor of both *lgs1* and incompatibility. The second parent was a high stimulant (*LGS1*) line with presumed susceptibility to *Striga*. Phenotyping included *Striga* germination stimulant activity, germination inhibition, haustorial initiation by the sorghum roots as well as the number of *Striga* attachments and whether these grew over the observation period. Although the second parent was *LGS1*, it did not contrast with the resistant parent sufficiently enough in terms of post-attachment success of *Striga* to distinguish the progeny lines for resistance at this stage. This co-culture method will be used to phenotype recombinant inbred lines derived from parents contrasting for incompatibility in efforts to identify gene(s) underlying this *Striga* resistance trait.



	Attachments @14d	Attachments @21d	Whole plate @21d
Striga susceptible at post-attachment phase			
Striga resistant at post-attachment phase			

Results

We were able to measure a variety of traits presumably contributing to *Striga* resistance in this sorghum population. These included low *Striga* germination stimulant activity by the maximum germination distance (MGD) and percent near-host *Striga* germination rate. Differences in MGD were overcome by spraying the agar with GR24. We also found some inhibitory effects of sorghum root exudates in several lines through the germination index (GI). We did not find much difference in haustorial initiation capacity among the sorghum lines, especially among those for which resistance was also measured in a pot study (Table 1). The number of parasitic attachments varied widely between lines. Attachments were marked and imaged at 14d and reimaged at 21d. Those that grew were assumed to have established vascular connections. All plants had some failed parasitic attachments. Among the 15 lines tested in *Striga* infested pots in Ethiopia, those supporting the least *Striga* biomass had, by our assay, either low *Striga* germination stimulant activity or had a low parasite success rate. A few had both. The assay made both measures possible. The presumed susceptible parent line P954063, although having high *Striga* germination stimulant, did show some post-attachment resistance. We will therefore phenotype a different recombinant inbred population along with bulked segregant whole genome sequencing to find candidate genes underpinning the incompatibility trait. The ability to track both *lgs1* and incompatibility by molecular markers will greatly assist breeding efforts to improve *Striga* resistance.

Table 1. *Striga* reaction measures from agar co-culture on 15 sorghum lines tested previously in *Striga* infested pots at Holeta, Ethiopia

Sorghum Line	PEDIGREE	Holeta GH 2021 *Striga* dry wt. at harvest (g)	Avg. no. *Striga* attachments/ plant @14 ± stddev	Avg. no. attachments that grew between 14 and 21 days ± stddev	% attached *Striga* that grew	% germ. *Striga* with haustoria	GI on avg. germ. rate of all plates	Avg. MGD ± stddev	Avg. *Striga* germ. rate before GR24 added
PSL985007; **P9405**	(SRN39 × P954063)F16	0.06	7.5 ± 5.1	1.8 ± 1.9	24%	81%	0.5	4.1 ± 3.1	20%
PSL985001; **P9402**	(SRN39 × P954063)F16	0.10	5.3 ± 2.5	0.3 ± 0.6	6%	77%	0.5	6.3 ± 2.3	20%
PSL985034	(SRN39 × P954063)F15	0.11	1.2 ± 1.8	0.5 ± 0.7	42%	84%	0.5	1.7 ± 1.1	6%
PSL984998; **P9401**	(SRN39 × P954063)F16	0.13	3.3 ± 3.3	0.5 ± 0.7	15%	82%	0.7	0.0 ± 0.0	0%
PSL985052	(SRN39 × P954063)F15	0.15	6.0 ± 2.6	0.3 ± 0.6	6%	76%	0.7	14.6 ± 2.9	56%
PSL985006	(SRN39 × P954063)F16	0.16	2.0 ± 1.7	0.2 ± 0.4	8%	90%	0.5	2.2 ± 3.0	4%
PSL985021	(SRN39 × P954063)F15	0.17	2.5 ± 0.7	0.5 ± 0.7	20%	89%	0.8	3.2 ± 0.7	19%
PSL985027	(SRN39 × P954063)F15	0.17	2.5 ± 3.1	0.7 ± 1.2	27%	80%	0.6	3.4 ± 2.2	9%
PSL985003; **P9403**	(SRN39 × P954063)F16	0.18	3.3 ± 1.5	0.0 ± 0.0	0%	86%	0.7	16.0 ± 3.0	50%
PSL985049	(SRN39 × P954063)F15	0.18	3.6 ± 1.7	0.2 ± 0.4	6%	83%	0.6	16.4 ± 7.4	47%
PSL985005; **P9404**	(SRN39 × P954063)F16	0.19	4.3 ± 0.6	0.7 ± 1.2	15%	83%	0.6	10.6 ± 5.7	40%
PSL985009; **P9406**	(SRN39 × P954063)F16	0.21	3.2 ± 2.6	0.5 ± 0.6	16%	80%	0.6	12.1 ± 1.3	45%
PSL985047	(SRN39 × P954063)F15	0.24	5.2 ± 1.5	1.2 ± 1.1	23%	70%	0.6	2.5 ± 1.7	12%
PSL984995; **P9407**	(SRN39 × P954063)F16	0.25	2.7 ± 1.2	0.7 ± 1.2	25%	85%	0.6	13.9 ± 9.0	46%
PSL985011; **P9408**	(SRN39 × P954063)F16	0.28	2.2 ± 1.8	0.4 ± 0.9	18%	85%	0.5	14.5 ± 5.1	39%
PRL984113	P954063	NT	3.3 ± 2.3	0.3 ± 0.5	10%	83%	0.8	18.5 ± 6.6	55%
PSL985429	SRN39	NT	3.6 ± 2.9	0.7 ± 1.1	20%	84%	0.6	0.4 ± 0.5	1%
	Averages among 15 tested lines	0.17	**3.7**	**0.6**	**17%**	**82%**	**0.6**	**8.1**	**27%**
	Range among all 60 lines		1 to 10	0 to 2.5	0 to 67%	69 to 95%	0.3 to 1.0	0 to 22.4	0 to 75%

Future Direction:

In the future we plan to continue the screening process on a much larger population from different parents but with similar genetics. These parents contrast for the incompatibility trait. The screening of the larger population will be done in hopes of identifying two very distinct groups that either possess or do not possess post-attachment *Striga* resistance. Once two distinct groups have been identified we plan use whole genome sequencing of bulked DNA from the susceptible and resistant individuals to hopefully identify the gene(s) responsible for post-attachment *Striga* resistance.

Background

The parasitic weed *Striga hermonthica* is a major production constraint to sorghum and other cereals in sub-Saharan Africa. *Striga* lacks its own root system and therefore must attach to its sorghum host and establish a vascular connection in order to survive to maturity. It can severely damage the sorghum crop under high infestations. *Striga* resistance is known in sorghum but only one gene, *LGS1* controlling resistance is known (1). Mutation at this locus reduces the *Striga* germination stimulant activity of sorghum root exudates. Other resistance mechanisms have been described but are difficult to measure, particularly in combination with *lgs1*, since this trait can mask subsequent resistance reactions, especially under field conditions (see figure on early steps in parasitic relationship). We screened a group of sorghum lines of shared pedigree derived from a known resistant line, SRN39, possessing both *lgs1* and an ill-defined post-attachment resistance mechanism called incompatibility. The objective was to find a relatively quick assay in which we can distinguish various mechanisms of *Striga* resistance with the ultimate goal of identifying the locus (loci) controlling incompatibility.

Materials and Methods:

Sorghum: SRN39, a *Striga* resistant line with low germination stimulant activity (*lgs1*) and incompatibility. P954063 a broadly adapted line with high germination stimulant activity (*LGS1*) and unknown post-attachment reaction to *Striga*. Recombinant inbreds derived from a cross between the two.

Striga: *S. hermonthica* seed collected from Sudan conditioned and embedded in 0.7% agar in 100×25mm Petri dishes.

Assay: [see (2) for complete details]. Agar-based co-culture in which the following non-destructive measures could be taken on the sorghum lines: 1) *Striga* germination stimulant activity (MGD - average of 3 furthest germinated *Striga* seeds from sorghum root and percent germination of *Striga* seed within a defined area of sorghum root), 2) *Striga* germination inhibition (GI - comparing *Striga* germination rate 2 days after adding GR24, a synthetic strigolactone, within 1cm of sorghum root with areas of plates distal from roots, 3) haustorial initiation capacity (% of germinated *Striga* within 1cm of sorghum root forming a haustorium), 4) number of *Striga* attachments on sorghum root at 14 days of co-culture, 5) number of *Striga* attachments that grew between 14 and 21 days.

Conclusions:

Our screening of the 58 inbred lines from the cross of two parents did not yield a large enough separation of individuals for post-attachment *Striga* resistance. Overall, there was poor separation for post attachment *Striga* resistance with the lowest being 0% and the highest being 67% but the lower and upper divisions did not contain enough individuals for bulk segregant analysis. This process did however confirm that *LGS1* is not responsible for post-attachment *Striga* resistance since it appears that post-attachment resistance is conferred from the other parent in the cross. This screening process also showed that our lab-based screening method of co-culturing *Striga* and sorghum to screen them for post-attachment resistance works.

References:

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2. Rich PJ. 2022. An agar based method for determining mechanisms of *Striga* resistance in sorghum. *In* Ghanim AMA, Sivasankar S, Rich PJ (eds) Mutation breeding in cereals for resistance to the parasitic weed *Striga*. Springer (in press).